

Voting Mechanisms, Welfare, and Equity in Participatory Budgeting

Presupuesto Participativo: An Approach from Durango, Mexico

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Motivation

Democracy Beyond the Ballot Box

Participatory Budgeting

(*Presupuesto Participativo*)

- Citizens vote on *how* to spend part of the public budget
- Born: Porto Alegre, Brazil (1989)
- Now in 11 000+ cities worldwide
- Mexico: adopted by many municipalities under *Leyes de Participación Ciudadana*

Why does it matter?

- Citizens gain a direct say in fiscal decisions
- Aligns government incentives with community needs
- Good allocation → political goodwill → accountability

Key insight: The outcomes depend critically on the *voting rules*.

Durango's *Presupuesto Participativo*: The Current Rules

The District-Exclusivity Rule (*Regla de Exclusividad Distrital*)

Each voter must select **one district** (*sector*) and allocate **all** votes within it. Voting across district boundaries is **prohibited**.

Mechanism in practice:

- n voters, each with $v = 6$ votes
- m districts in Durango
- Projects registered within each district
- Top vote-getters funded greedily until budget exhausted

Stated justification:

- Prevent high-visibility city-center projects from absorbing all votes
- Protect marginalized *colonias* from being shut out
- Ensure geographic representation

The Central Question

Does the district-exclusivity rule achieve its goals?

And if not — what should we use instead?

This paper:

1. Formally identifies **five distortions** caused by the current rule
2. Shows that simply removing the constraint **destroys equity**
3. Proposes the **Method of Equal Shares** (*Método de Partes Iguales*) as a reform
4. Characterizes when and why MES achieves both efficiency *and* equity

A Brief Framework: Mechanism Design

What Is Mechanism Design?

“Engineering the rules of the game so that self-interested agents produce good social outcomes.”

(1) Environment

Who are the players?

What do they want?

What are the constraints?

(2) Social Choice Function

What outcome *should* we produce given everyone's preferences?

(3) Mechanism

What *rules* (message space + outcome function) produce that outcome?

The design problem

Preferences are *private*. How do we design rules so that agents reveal them truthfully?

Key Definitions

Environment $\langle N, X, \Theta, u \rangle$

n agents with private types $\theta_i \in \Theta_i$ (preferences + information); utility functions $u_i : X \times \Theta_i \rightarrow \mathbb{R}$.

Social Choice Function $f : \Theta \rightarrow X$

The planner's ideal rule: maps type profiles to outcomes. *Example*: fund the set of projects with highest total welfare.

Mechanism $(\mathcal{M}, g)$

$\mathcal{M} = \prod_i M_i$ is the **message space** (what agents report); $g : \mathcal{M} \rightarrow X$ is the **outcome function** (how messages map to results).

The Revelation Principle

Theorem (Revelation Principle)

*If a social choice function f can be implemented by any mechanism in Nash equilibrium, it can also be implemented by a **direct, incentive-compatible** mechanism — one where truthful reporting is a dominant strategy.*

Implication for us: We only need to study *direct* mechanisms (voters report preferences directly). Restricting to truthful reporting is without loss of generality.

The Gibbard-Satterthwaite Impossibility

Theorem (Gibbard 1973; Satterthwaite 1975)

*With three or more alternatives and unrestricted preferences, any **strategy-proof** social choice function that can select any alternative must be **dictatorial**.*

- No non-dictatorial mechanism is fully strategy-proof in a general setting
- The **goal shifts**: design mechanisms that are *as good as possible* under this constraint
- We will revisit this when we evaluate the Method of Equal Shares

Both Durango's current rule *and* the proposed alternative fail strategy-proofness. The difference lies in everything else.

The Model

The Participatory Budgeting Environment

Definition: PB Environment $\langle N, \mathcal{D}, \mathcal{P}, B, v, u, \sigma \rangle$

- $N = \{1, \dots, n\}$: voters
- $\mathcal{D} = \{1, \dots, m\}$: districts (*sectores*)
- $\mathcal{P} = \bigsqcup_d \mathcal{P}^d$: projects, partitioned by district
- $c : \mathcal{P} \rightarrow \mathbb{R}_{>0}$: cost function
- $B > 0$: total budget
- $v \in \mathbb{N}$: vote endowment per voter
- $u_i(p) \geq 0$: voter i 's utility for project p
- $\sigma_i(p) \in \{0, 1\}$: **information function**
“Does voter i know about project p ?”

Why σ_i ?

Voters know their neighborhood's projects — not the whole city's. This **rational ignorance** (Downs, 1957) is the root cause of several distortions.

District Salience (*Visibilidad Distrital*)

Definition: Salience α_d

Fraction of voters informed about *at least one* project in district d :

$$\alpha_d = \frac{|\{i \in N : \exists p \in \mathcal{P}^d, \sigma_i(p) = 1\}|}{n}$$

High-salience ($\alpha_d \approx 1$):

City center, university district.

Most voters know the projects.

Low-salience ($\alpha_d \approx 0$):

Peripheral, marginalized *colonias*.

Few voters know the projects.

Salience drives both **strategic behavior** (who goes where) and **equity outcomes** (who gets funded).

Two Mechanisms: DEM and UAM

District-Exclusivity Mechanism (DEM)

- Voter selects **one district** $d_i \in \mathcal{D}$
- Allocates all v votes *within* $\mathcal{P}^{d_i}$
- **Durango's current rule**

Unrestricted Allocation Mechanism (UAM)

- Allocates v votes across **any** project in **any** district
- No geographic constraint
- The “obvious” alternative

Both mechanisms use the same **greedy outcome rule**: fund projects in decreasing vote order until the budget is exhausted.

$$\mathcal{M}^{\text{DEM}} \subsetneq \mathcal{M}^{\text{UAM}}$$

Every DEM vote profile is a valid UAM profile. The DEM simply *restricts* the feasible space.

Five Distortions of the Current Mechanism

Distortion 1: Preference Truncation (*Truncamiento de Preferencias*)

Proposition

Under sincere voting *with full information*, utilitarian social welfare satisfies $W^{\text{DEM}} \leq W^{\text{UAM}}$, with strict inequality whenever the marginal voter has cross-district preferences.
(*With incomplete information the ranking can reverse — this is exactly Distortion 5.*)

Intuition:

- A voter who cares about projects in *two* districts must abandon half their preferences
- All 6 votes are forced into one district — even if the voter's top priority lies elsewhere
- The DEM *artificially narrows* each voter's expressed preference profile
- Since $\mathcal{M}^{\text{DEM}} \subsetneq \mathcal{M}^{\text{UAM}}$, the DEM truncates the preferences each voter can express

Distortion 2: Noise Injection (*Inyección de Ruido*)

Noise Allocation Model

A voter selecting district d must allocate *all* v votes within $\mathcal{P}^d$. Votes on projects the voter doesn't know become **noise**: random allocation unrelated to quality.

Proposition

For any project $p \in \mathcal{P}^d$:

$$\mathbb{E}[V(p)] = \underbrace{S(p)}_{\text{signal: votes from informed voters}} + \underbrace{\mathbb{E}[\eta(p)]}_{\text{noise: votes from uninformed voters}}$$

$\mathbb{E}[\eta(p)]$ is **independent of** $u_j(p)$ for all voters j . The noise-to-signal ratio rises with the number of unknown projects.

Noise in Action: Durango-Scale Example

Setup: District A, 8 projects, 100 voters, $v = 6$

- 40 institutional voters: know only p_1 (flagship project)
- 30 local residents: know 3 projects including p_1
- 30 mobilized supporters: each knows one other project (not p_1)

Dominant project p_1 :

Signal $S(p_1) \approx 220$

Noise $\mathbb{E}[\eta(p_1)] \approx 4.3$

Noise/Signal $\approx 2\%$ ✓

Small valued project p_7 :

Signal $S(p_7) = 10$

Noise $\mathbb{E}[\eta(p_7)] \approx 15.7$

Noise/Signal $\approx 157\%$ ✗

For marginal projects, noise *exceeds* the signal. Funding at the margin is driven by random allocation, not genuine preferences.

Distortion 3: The Coordination Game (*Juego de Coordinación*)

District-Selection Game Γ

Voters with positive utility in multiple districts face a strategic choice: which district to select? Payoffs depend on others' choices — creating **strategic complementarity**.

Proposition

If enough voters have cross-district preferences, Γ has **at least two Nash equilibria** producing **distinct funded sets** with **distinct welfare levels**.

Example: Districts A and B each have one borderline project; 50 mobile voters can tip either.

- E_A : all mobile voters $\rightarrow A$ — p_A funded, p_B not
- E_B : all mobile voters $\rightarrow B$ — p_B funded, p_A not

Which project gets funded depends on *who shouts louder*, not on what the community values.

Distortion 4: Mobilization Advantage (*Ventaja por Movilización*)

Proposition

A mobilizing institution $(\mathcal{C}, p_{\mathcal{C}})$ coordinating $|\mathcal{C}|$ voters:

- (i) Inflates district $d_{\mathcal{C}}$'s vote pool by $|\mathcal{C}| \cdot v$
- (ii) Creates vote displacement $\Delta_{\mathcal{C}} = v \cdot |\mathcal{C}^*|$ ($\mathcal{C}^*$: members diverted from their preferred district)
- (iii) Causes private utility loss to diverted members, $\mathcal{L}(\mathcal{C}) \geq 0$

Example: A university coordinates 400 employees $\rightarrow$ the district with the university park project.

- $\Delta_{\mathcal{C}} = 400 \times 6 = 2,400$ displaced votes
- Those 400 voters can no longer support the projects they actually value most
- The outcome reflects institutional power, not community preferences

Distortion 5: The Equity Tradeoff (*Dilema de Equidad*)

Proposition

- **UAM**: High-salience districts capture nearly all votes. Marginalized, low-salience districts may get *zero funding* (β -equity fails with $\beta = 0$).
- **DEM**: Committed local residents create a *protected vote pool*, giving marginalized districts partial protection ($\beta > 0$).

The core dilemma

The DEM's district constraint is a **blunt** equity instrument.

It partially protects marginalized districts — but at the cost of distortions 1–4.

Simply removing the constraint (UAM) makes equity worse.

We need a mechanism that achieves equity *by design*, not by restriction.

The Method of Equal Shares

The Core Idea: Individual Budget Shares

Each voter receives an **equal share** of the budget:

$$b_i = B/n$$

Projects are funded by **pooling** the shares of their supporters.

What voters do:

- Submit an **approval ballot** $A_i \subseteq \mathcal{P}$
- Approve any projects they support,
across any district
- Only approve projects they know:
 $\sigma_i(p) = 1$

What the algorithm does:

- Finds the project with the *most support relative to cost*
- Deducts from supporters' shares proportionally
- Repeats until no more projects are affordable

The MES Algorithm

Definition: Method of Equal Shares

Initialize: $b_i = B/n$ for all i ; $W = \emptyset$.

Repeat until no project is affordable:

1. For each $p \notin W$, find supporters $N(p) = \{i : p \in A_i, b_i > 0\}$
2. Compute the **cost-per-supporter threshold**:

$$\varrho(p) = \min \left\{ \varrho \geq 0 : \sum_{i \in N(p)} \min(b_i, \varrho) \geq c(p) \right\}$$

3. **Fund** $p^* = \arg \min_p \varrho(p)$; deduct $\min(b_i, \varrho(p^*))$ from each $i \in N(p^*)$

Completion: Allocate remaining budget greedily by total approval count.

MES Step-by-Step: Example

4 voters, $B = 100$, $b_i = 25$ each

Project	Cost	District	$i=1$	$i=2$	$i=3$	$i=4$
p_1 : Park renovation	40	Center	✓	✓	✓	
p_2 : School repair	30	Marginal			✓	✓
p_3 : Road paving	50	Center	✓	✓		
p_4 : Water access	20	Marginal				✓

Round 1 — thresholds: $\varrho(p_1) = 13.3$, $\varrho(p_2) = 15$, $\varrho(p_3) = 25$, $\varrho(p_4) = 20$.

→ **Fund p_1** . Deduct 13.3 from $i = 1, 2, 3$. Remaining: $b_{1,2,3} = 11.67$; $b_4 = 25$.

MES Step-by-Step: Example (cont.)

After Round 1: $b_1 = b_2 = b_3 = 11.67$; $b_4 = 25$

Round 2 — p_2 : $\varrho(p_2) = 18.3$; p_4 : $\varrho(p_4) = 20$.

p_3 : total available = $2 \times 11.67 = 23.3 < 50$ — *not affordable*.

→ Fund p_2 (smallest ϱ). $i = 3$ pays 11.67, $i = 4$ pays 18.33. Now $b_3 = 0$; $b_4 = 6.67$.

Round 3 — p_4 : only $i = 4$ left, $b_4 = 6.67 < 20$ — not affordable.

p_3 : not affordable. **Stop.**

Completion: Remaining budget = 30. Fund p_4 (cost 20) greedily. Final: $W = \{p_1, p_2, p_4\}$.

The “Marginal” district gets two projects — without any geographic constraint.
Budget proportionality did the work that the DEM tried to do by force.

Extended Justified Representation (EJR)

Definition: EJR

A funded set W satisfies **Extended Justified Representation** if: for every cohesive group S of voters whose collective budget share covers their projects, at least one member of S has their proportional share of the budget represented in W .

Theorem (MES Satisfies EJR (Peters et al., 2021))

The Method of Equal Shares always produces outcomes satisfying EJR.

What this means for Durango: Any community of residents who collectively have enough budget share to fund their projects *will* have those interests represented in the outcome — by construction, not by chance.

MES Resolves All Four Distortions

Prop. 6: No Preference Truncation

Ballot space $M_i^{\text{MES}} = 2^{\mathcal{P}}$: approve anything, across any district. No cross-district restriction.

Prop. 7: No Forced Noise

Voters approve only projects they know. No minimum ballot size. Noise votes do not arise.

Prop. 8: No Coordination Game

No district-selection stage. The strategic complementarity that creates multiple equilibria is absent by design.

Prop. 9: Bounded Mobilization

Coalition $\mathcal{C}$ can direct at most $|\mathcal{C}| \cdot (B/n)$ to any project. Vote displacement $\Delta_{\mathcal{C}} = 0$.

MES and Geographic Equity (*Equidad Geográfica*)

Proposition (Geographic Equity via EJR)

Let $\beta_d = n_d/n$ be district d 's population share. If the committed residents of district d *commonly approve* a local bundle worth $\beta_d B$, then under MES the funded set W satisfies:

$$\sum_{p \in W \cap \mathcal{P}^d} c(p) \geq \beta_d \cdot B.$$

- Every such district gets **at least its proportional budget share** — guaranteed
- This is **structural** (follows from EJR): holds regardless of what other voters do
- Contrast with DEM: equity depends on committed residents happening to form a large enough bloc
- If local preferences are heterogeneous, the floor is the largest *commonly-approved* bundle — still > 0 , still beats the UAM's $\beta = 0$

An Honest Limitation: MES Is Not Strategy-Proof

Proposition

MES does not satisfy DSIC: there exist preference profiles where a voter gains by misreporting their ballot.

- **Inherent, not incidental:** Peters et al. (2021) (Thm. 4.8) prove that no EJR-satisfying mechanism can be strategy-proof
- The tradeoff between *proportional representation* and *strategy-proofness* cannot be resolved
- MES is strategically *simpler* than the DEM: no district selection game, no coordination failures, no mobilization arms race

The DEM also fails strategy-proofness (its district-selection stage is directly manipulable). Both mechanisms lose on the same criterion — MES wins on everything else.

In practice the manipulation needs detailed knowledge of others' ballots and is risky — low-stakes at scale.

Two More Honest Limitations (*Otras Limitaciones*)

The completion gap

MES often **stops before spending the whole budget**: once no project is affordable to its backers, a *completion* step (greedy by approvals) places the rest — with **none** of the EJR guarantee.

- The gap depends on project costs vs. the per-voter share $b_i = B/n$: many small projects → the proportional phase reaches far; few expensive projects → completion dominates
- **Design lever**: keep project cost tiers small relative to B/n

Ballot complexity

MES asks voters to consider *all* projects city-wide, not just one district's.

- With 50+ projects this is a real cognitive load; thin ballots waste a voter's share
- **Mitigation**: a grouped ballot interface with a live “budget impact” indicator — richer than the DEM, still tractable

Comparing the Three Mechanisms

The Property Table

Property	DEM	UAM	MES
No preference truncation	✗	✓	✓
No forced noise injection	✗	✓	✓
No coordination game	✗	✓	✓
Bounded mobilization power	✗	Partial	✓
β -geographic equity	Partial	✗	✓
Strategy-proof (DSIC)	✗	✗	✗
Proportional representation	✗	✗	✓

MES strictly dominates both alternatives on every property except strategy-proofness — which no proportional (EJR) mechanism can satisfy (Peters et al., 2021).

300-Voter Comparison: Setup (*Escenario a Escala Durango*)

$n = 300$ voters, $B = 300$, $v = 6$

	Committed residents	Salience α_d	Projects
District A (Center)	100	0.90	p_1 (cost 120), p_2 (cost 80)
District B (Residential)	120	0.50	p_3 (cost 100)
District C (Marginalized)	80	0.15	p_4 (cost 60), p_5 (cost 40)

Additional types: 80 mobile voters in B also approve p_1 ; 20 connected voters in C also approve p_1 .

300-Voter Comparison: Results

	DEM	UAM	MES
District <i>A</i> funded	p_1	p_1, p_2	p_1
District <i>B</i> funded	p_3	p_3	p_3
District <i>C</i> funded	p_4	nothing	p_5
Noise votes present	Yes	No	No
Coordination required	Yes	No	No
Vote displacement	70 voters	0	0

- UAM: excludes the marginalized district entirely despite genuine demand
- DEM: partially protects district *C* but wastes 70 voters' preferences
- MES: funds district *C* through proportionality alone — **no constraint needed**

Conclusion and Policy Implications

Policy Recommendations (*Recomendaciones de Política*)

1. **Replace the DEM with MES in Durango's *presupuesto participativo***
 - Allow cross-district approval voting
 - Geographic equity is achieved structurally, not by restriction
 - Implementation is straightforward: voters mark projects they support
2. **Invest in project-level information campaigns**
 - MES rewards informed voters: more $\sigma_i(p) = 1$ entries $\rightarrow$ better outcomes
 - Targeted publicity for projects in low-salience, marginalized districts

Counterfactual Simulation: Durango 2025

Setup: 48 projects, 6 districts, \$500,000 MXN budget. Synthetic approval ballots ($n = 1,000$, seed 42) generated from observed 2025 vote totals.

Metric	DEM (observed)	MES (counterfactual)
Welfare (total votes on funded projects)	4,985	4,888
Gini (district-level equity)	0.300	0.367
Projects funded	10	10
District 1 funded (MXN)	50,000	100,000
District 2 funded (MXN)	150,000	150,000
District 3 funded (MXN)	100,000	50,000
District 4 funded (MXN)	0	0
District 5 funded (MXN)	100,000	50,000
District 6 funded (MXN)	100,000	150,000

Key finding

MES applied to *DEM-constrained* ballots does not outperform DEM. Since synthetic voters each select only one district, MES never observes cross-district preferences — the same information gap that drives Propositions 1 and 2. The simulation **illustrates the distortion**, not MES's full potential.

Summary (*Resumen*)

1. The **District-Exclusivity Mechanism** introduces **five distortions**: preference truncation, noise injection, coordination failures, mobilization advantage, and an equity tradeoff
2. The “obvious” fix (**UAM**) **destroys geographic equity** for marginalized districts
3. The **Method of Equal Shares** resolves all five distortions while providing **structural** geographic equity via EJR
4. A counterfactual simulation on Durango 2025 data shows that MES applied to DEM-constrained ballots **replicates the distortion** — confirming that full welfare gains require reforming *both* the mechanism and the ballot format

*Democracy is not only cast in the ballots.
It is also in the rules that translate ballots into outcomes.*

Thank you

Gracias

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